

TivaC-general-info

Model/Chips names

Middle chip serials: TMC41230 H6PMT7 15075VW 64

Top chip serials: 980 TM4C123G H6PM 11A5TTW G4

Model name:

- Tiva C Series TM4C123G
- LaunchPad Evaluation Kit
- EK-TM4C123GXL

Back of the box info

Tiva™M C Series 64-pin TM4C123GH6PMI digital signal controller

- 80MHz 32-bit ARM® Cortex™-M4 based MCU with floating point
- 256KB Flash /32KB SRAM/2KB EEPROM on-chip memory
- Dual 12-bit ADC, 6x64-bit and 6x32-bit timers, MC PWMs, many serial comms

Download extensive TivaWare™ C Series code base

- FREE example applications
- FREE Device driver library
- FREE Graphics library
- FREE USB library
- ...and more!

Select and download upgradeable development tool chains

- Code Composer Studio™ IDE: full functional board locked
- Keil™: 32KB address limited
- IAR: 32KB address limited
- Mentor Embedded Sourcery™ CodeBench: 30-day full functional

Included in the box!

- Tiva™M C Series LaunchPad evaluation board (EK-TM4C123GXL with MCU)
- USB debug cable
- Read Me First instructions

Launchpad Workshop "Interesting" Information

JTW and Serial Wire Debug debugger access

- ETM (Embedded Trace Macrocell) available through Keil and IAR emulators
- 256KB Flash memory
- 2KB EEPROM (fast, saves board space)
- 32KB single-cycle SRAM with bit-banding

Battery-backed Hibernation Module

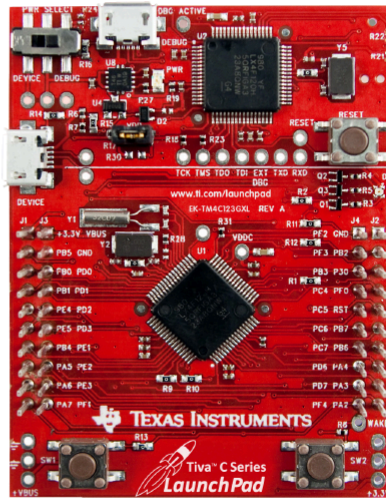
- Internal and external power control (through external voltage regulator)
- Separate real-time clock (RTC) and power source
- VDD3ON mode retains GPIO states and settings
- Wake on RTC or Wake pin
- Sixteen 32-bit words of battery backed memory
- 5 µA Hibernate current with GPIO retention. 1.7 µA without

Serial Connectivity

- USB 2.0 (OTG/Host/Device)
- 8 - UART with IrDA, 9-bit and ISO7816 support
- 6 - I2C
- 4 - SPI, Microwire or TI synchronous serial interfaces
- 2 - CAN

Tiva™ EK-TM4C123GXL LaunchPad

- ◆ ARM® Cortex™-M4F
64-pin 80MHz TM4C123GH6PM
- ◆ On-board USB ICDI
(In-Circuit Debug Interface)
- ◆ Micro AB USB port
- ◆ Device/ICDI power switch
- ◆ BoosterPack XL pinout also supports
legacy BoosterPack pinout
- ◆ 2 user pushbuttons
(SW2 is connected to the WAKE pin)
- ◆ Reset button
- ◆ 3 user LEDs (1 tri-color device)
- ◆ Current measurement test points
- ◆ 16MHz Main Oscillator crystal
- ◆ 32kHz Real Time Clock crystal
- ◆ 3.3V regulator
- ◆ Support for multiple IDEs:



mentor
embedded

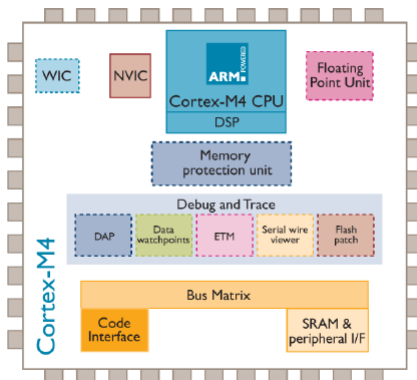
UIAR
SYSTEMS

ARM KEIL
M4

the Composer
Studio

Lab...

A 32 or 64-bit Windows XP, Windows7 or 8 laptop with 2G or more of free hard drive space. 1G of RAM should be considered a minimum ... more is better.
Apple laptops running any of the above OS's are acceptable. **Linux laptops are not recommended.**



- **Tiva C Series TM4C123G LaunchPad Evaluation Board**
 - Tiva TM4C123GH6PMI microcontroller
 - On-board In-Circuit Debug Interface (ICDI)
 - Supply voltage: 4.75 ~ 5.25 VDC
 - USB micro-A and micro-B connectors for USB device, host, and on-the-go (OTG) connectivity
 - RGB user LED
 - Two user switches (application/wake)
 - Available I/O brought out to headers on a 0.1-in (2.54-mm) grid
 - Switch-selectable power sources: ICDI – USB
 - Reset switch

Board = system

MCU = brain

I/O = senses